AP BIOLOGY · UNIT 7 · NATURAL SELECTION

Allele Machine

What dominance does and does not control, what a null model is for, and why a single drifting population is unpredictable while two hundred of them are not.

Name ______________________ Date ______________________ Period ______________________

Demonstration: andresdbp.github.io/resources/demos/allele-machine/

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1 Before you open anything

Answer from what you already know. The point is to have a number on paper that the demonstration can later confirm or contradict.

PREDICTION

In a large population, allele **A** is *completely dominant* over **a** and currently sits at a frequency of **p = 0.02**. Mating is random, no one has a survival or reproductive advantage, nobody arrives or leaves, and no mutation occurs. After **200 generations**, what is p?

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Write one or two sentences saying how you got that number — in particular, say what role the word *dominant* played in your reasoning.

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2 Reading the machine

Open the demonstration and stay on the **Hardy–Weinberg** tab. This tab models an infinite population with exact arithmetic, so nothing here is noisy: any movement in p is a force you switched on, not a fluke.

1. Press **Restore all five conditions** and check that the second panel now reads **Break a condition — 0 of 5 broken**.
2. Set **Dominance of A (h)** to **1.00** and **Generations** to **60**.
3. Set **Starting frequency of A (p)** to **0.30**. Fill in the first row from the readouts: **p (freq. of A)**, **Change in p**, and the three percentages listed under **Genotype frequencies**.
4. Repeat for each starting p in the left-hand column.

Starting p	p after 60 generations	Change in p	AA	Aa	aa
0.30					
0.50					
0.80					

a. For the **0.80** row, calculate p^2, $2pq$ and q^2 by hand. Write the three numbers and say which recorded column each one matches.

b. Every row has the same value in the **Change in p** column. What is it, and what does that value tell you about a population in which all five conditions hold?

c. Try to find a starting p that pushes the heterozygote frequency above 50%. Record the highest Aa value you can reach and the p that produced it. Then justify that maximum algebraically from 2pq.

3 What dominance actually controls

One control varied, everything else held still. Watch which readouts respond and which do not.

1. Press **Rare dominant allele**. This sets p to **0.02** and h to **1.00**, and restores all five conditions — exactly the population described in your part 1 prediction.
2. Set **Generations** to **200**.
3. Fill in the first row. The last column asks for the category *names* printed under **What you would actually see** — copy them, not their numbers.
4. Now move **Dominance of A (h)** to each value in the left-hand column, changing nothing else, and complete the table.

h	p after 200 generations	Change in p	Aa	Categories under “What you would actually see”
1.00				
0.75				
0.50				
0.25				
0.00				

a. Which columns changed as h moved, and which did not?

b. State the rule in your own words: dominance is a statement about __________, and it determines __________ but not __________.

c. Go back to your part 1 prediction. What did the demonstration give for p after 200 generations? If your prediction differed, name the specific assumption that did not hold.

> Dominance has one further job, and the demonstration is explicit about it: switch on **Natural selection** and h also sets the heterozygote’s relative fitness, which is why a recessive allele is so slow to remove at low frequency. On its own, with selection off, h still changes nothing about the gene pool.

4 Working backwards from a phenotype

Exam questions rarely hand you p. They hand you the one class of people you can count — the ones showing the recessive trait — and expect you to work back to the gene pool.

1. Press **Restore all five conditions**. Set **Dominance of A (h)** to **1.00**, **Starting frequency of A (p)** to **0.60**, and **Generations** to **60**.
2. Fill in the first row: the three **Genotype frequencies**, the **a trait (aa only)** percentage from **What you would actually see**, and the **Change in p** and **q (freq. of a)** readouts.

3. Now tick **Non-random mating** and set the **Inbreeding coefficient F** to **0.50**. Record the second row.

Setting	AA	Aa	aa	a trait (aa only)	Change in p	q (freq. of a)
F = 0 (random)						
F = 0.50						

a. Using the first row only: of the individuals showing the *dominant* trait, what fraction are heterozygous? Show the division you used.

b. Explain why you cannot get p directly from the frequency of the dominant phenotype, but you can from the frequency of the recessive one.

c. Now take the *second* row and apply the usual shortcut: q = √(frequency of aa). What value of q does that give? Compare it with the **q (freq. of a)** readout on screen.

d. The **Change in p** readout is identical in both rows, yet all three genotype frequencies moved. Reconcile those two facts, and state the condition the √ shortcut silently assumes.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it, set the controls the problem describes, and check yourself — if a number is off, write down where.

1. A population meets all five Hardy–Weinberg conditions, and 9% of individuals show the recessive phenotype. Find q, p, and the frequencies of AA, Aa and aa. Then find what fraction of the individuals showing the dominant phenotype are carriers.

2. Two hundred isolated populations of $N = 25$ each start at p = 0.20 and are left to drift, with no selection, mutation or migration. **(i)** In the long run, roughly how many of the 200 end up fixed for A, and what reasoning gives that number? **(ii)** Using $H_t = H_0(1 - 1/2N)^t$, predict the mean heterozygosity after 100 generations.

3. Two populations both start at p = 0.50 with no selection: one has $N = 20$, the other $N = 1000$. Calculate the expected mean heterozygosity in each after 200 generations. Then state, in one sentence, what the pair of numbers says about the claim that drift only matters in small populations.

4. No calculation for this one. A lab report says a population “returned to Hardy–Weinberg equilibrium” after a disturbance. Explain what is wrong with treating Hardy–Weinberg as something a population *does*, and say what information you actually get from measuring how far a real population sits from the Hardy–Weinberg proportions.

6 On your own

Nothing here is graded. The demonstration does considerably more than this worksheet uses.

- Break **Natural selection** on its own with s = 0.50 over 200 generations, and run it twice: once with h = 1.00 and once with h = 0.00. Selection is acting against the same allele at the same strength both times. Time how long the last few percent take in each case, and account for the difference using the genotype bands.
- Break **Migration (gene flow)** with m = 0.10 and a source frequency of 0.90. Check whether the remaining gap between p and 0.90 really shrinks to (1 - m) of itself every generation. Then compare how far **Mutation** can move p at its largest available rate over the same span.
- Break **Finite population size** alone at N = 10 and press **Draw again** several times, then move to the **Genetic drift** tab and press **New replicates** a few times. Watch what stays stable across presses (the **Mean p, all 200** readout, the fraction fixed) and what does not.